

# Project 2

## L.EIC – IART

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# Customer

- Our customer is a Real Estate Agency that wants to optimize the process of evaluating a house price within the Porto district.
- The product will also be used by individual people interested in easily knowing how much their house is worth



# Problem

- Currently there is no simple way to figure out the market value of a home. When a person wants to put their house in the market they usually have to pay a professional to evaluate its market value.
- This process is not immediate and easily accessible, besides also costing time and money to the customer.



# Available data

- To train our model, we imported a Kaggle dataset containing over 135,000 real estate listings with 25 structural, geographic, and quantitative attributes.
- This dataset captures property location, spatial dimensions (living/total area), and property metadata (construction year, room counts).
- Narrowed the dataset scope strictly to houses and apartments located within the Porto district to align with project goals.

# Solution overview

- For this, a web app was developed, using JavaScript, HTML and CSS at the frontend while Python handles the backend. It was designed to deliver instant, real-time property price for users.
- Powered by a high-performance Stacking Regressor that intelligently combines predictions from Linear Regression, Random Forest, and XGBoost models.

The screenshot shows a web browser window with the URL `http://127.0.0.1:5000`. The page title is "Porto House Value Estimator" and it includes the subtitle "Trained on 11 184 filtered Porto properties." The form is organized into several sections:

- LOCATION**: Includes "Municipality" (set to "Amarante") and "Parish" (set to "Aboadela, Sanche e Várzea").
- PROPERTY**: Includes "Type" (set to "Apartment"), "Energy certificate" (set to "A"), and "Conservation status" (set to "Good condition").
- BUILDING FEATURES**: Includes "Parking" (set to "0.0"), "Elevator" (set to "False"), and "Floor" (set to "1st Floor").
- AREAS AND ROOMS**: Includes "Total area (m2)" (set to "96"), "Living area (m2)" (set to "92"), "Construction year" (set to "1995"), "Total rooms" (set to "3"), "Bedrooms" (set to "3"), and "Bathrooms" (set to "2").

Each input field has a small text below it indicating the observed and common ranges for that feature. For example, for "Total area (m2)", it says "Observed range 0-274; common range 33-237. Larger values are allowed." For "Construction year", it says "Observed range 1900-2025; common range 1937-2004. Larger values are allowed."

On the right side of the form, there is a box titled "PREDICTED PROPERTY VALUE" showing the result: **EUR 169,216.75**. Below this, it says "Estimate for Aboadela, Sanche e Várzea, Amarante." and "Model: StackingRegressor". At the bottom of this box, it shows "Training region: Porto district" and "Notebook R2: 0.8641".

# Predictive model

**Target variable:** Price

## Numerical attributes

- TotalArea
- LivingArea
- GrossArea
- BuiltArea
- LotSize
- TotalRooms
- NumberOfBedrooms
- NumberOfBathrooms
- NumberOfWC
- ConstructionYear

## Categorical attributes

- District
- Municipality
- Parish
- Type
- Floor
- ConservationStatus
- EnergyCertificate
- Parking
- Garage
- Elevator
- ElectricCarsCharging
- PublishDate

- Categorical variables are converted via **One-Hot Encoding** to create binary inputs.

# Empirical study: setup

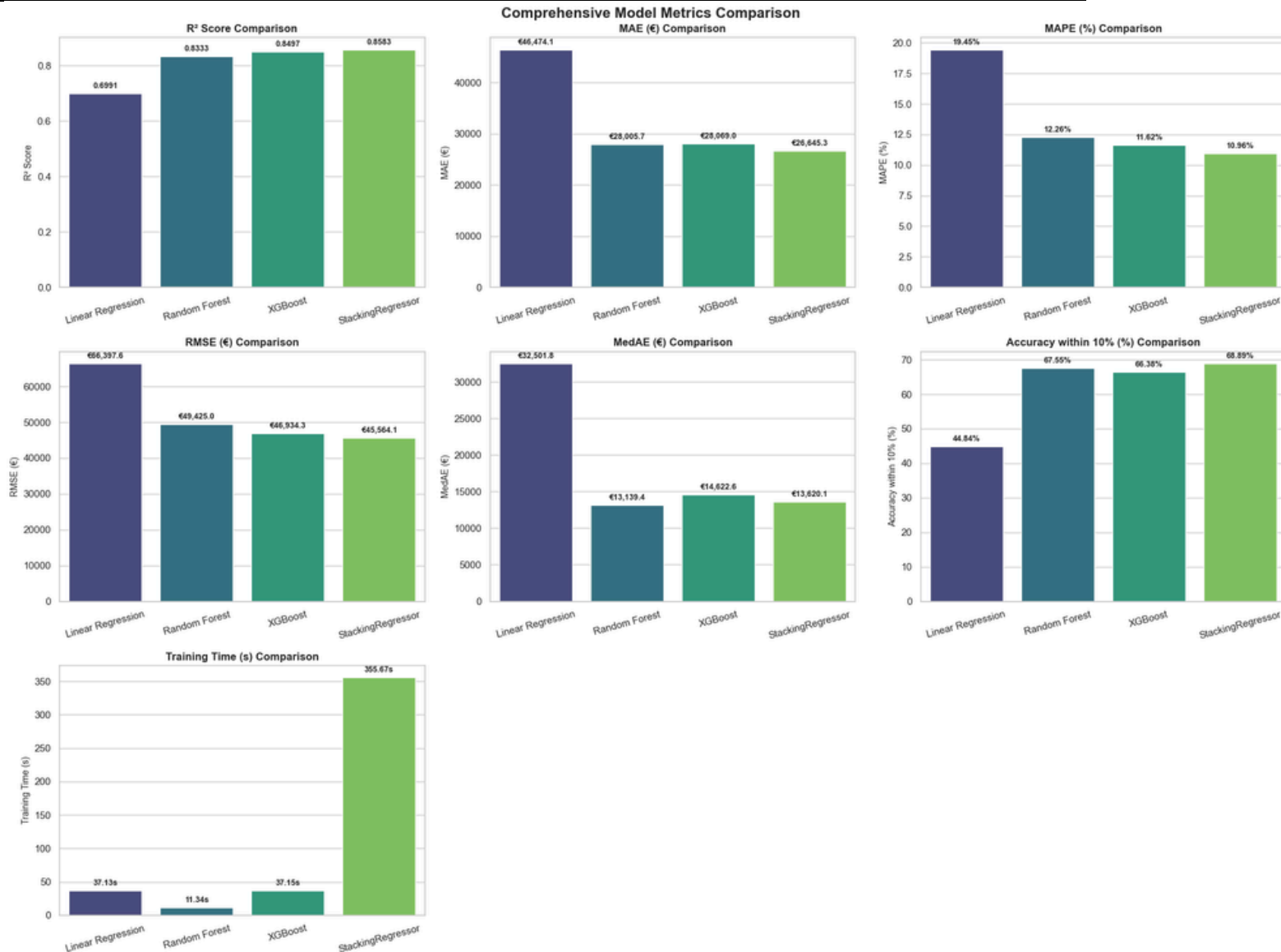
## Compared 4 regression algorithms

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- Stacking Regressor

## Evaluation metrics:

- MAE
- MAPE
- RMSE
- $R^2$  score
- MedAE
- Accuracy within 10%
- Training time

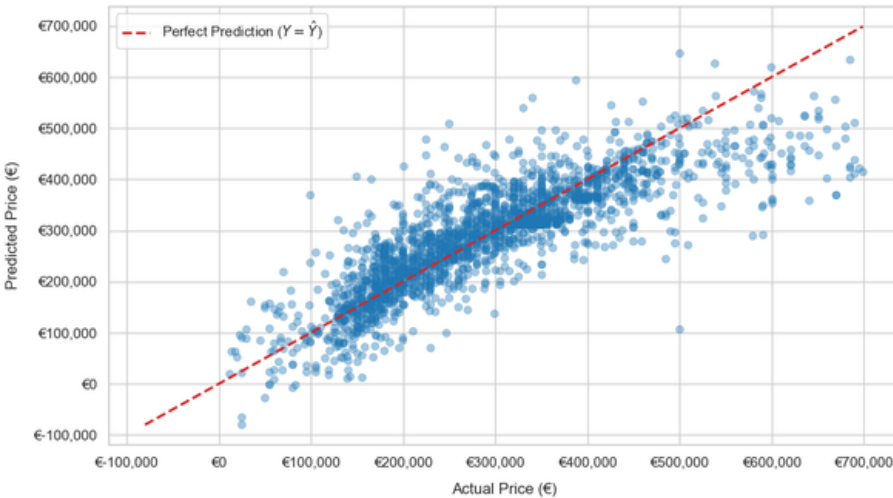
# Empirical study: results



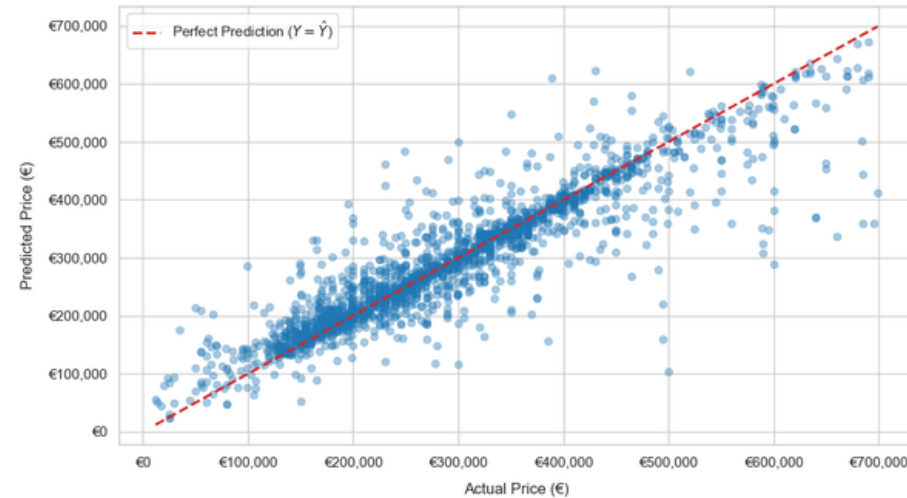


# Empirical study: results

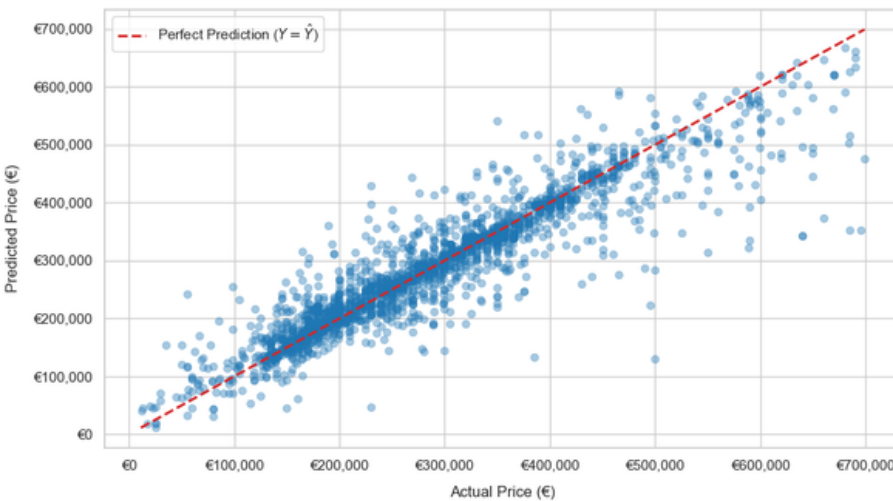
Actual vs. Predicted Property Prices (Linear Regression)



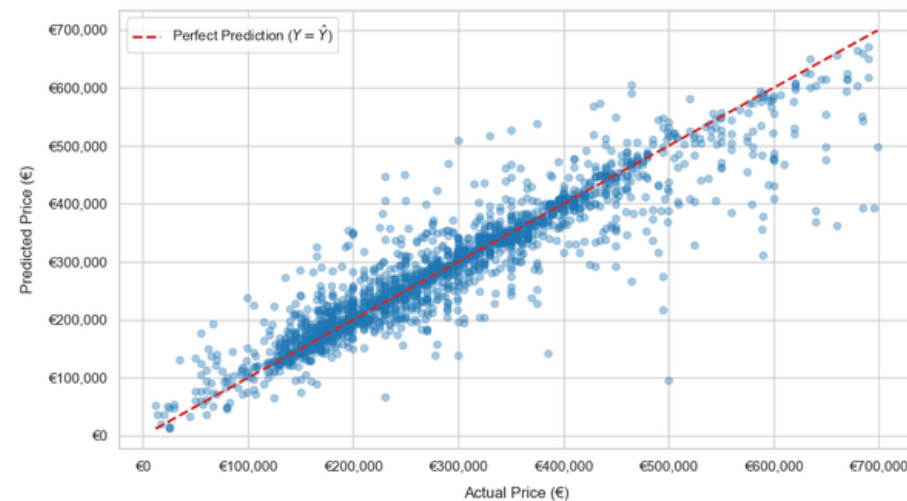
Actual vs. Predicted Property Prices (Random Forest)



Actual vs. Predicted Property Prices (XGBoost)



Actual vs. Predicted Property Prices (StackingRegressor)



# Stakeholders

- End-Users (Buyers & Sellers)
- Real Estate Professionals
- Investors & Developers

# Expected challenges during development and deployment

- **Initial Learning Curve:** First time architecting and configuring complex machine learning models, requiring significant experimentation with ensemble stacking methods.
- **Anticipated Data Gaps:** Expected critical structural features and localized market metrics to be missing, forcing reliance on strict filtering and imputation techniques.

# Unavailable data

- **Missing Market Drivers:** Lacks hyper-local economic data (e.g., proximity to public transit, school ratings, crime rates, or grocery hubs).
- **Property-Specific Qualities:** Missing precise interior condition metrics, such as recent renovation history, quality of finishes, view type (e.g., city vs. river view), or building orientation/sun exposure.
- **Missing values:** Due to missing values on row cells, we had to eliminate certain cells or apply an average value for the model to properly train.

# Ethical risks

- **Risk of Market Distortion:** Widespread reliance on a single valuation app could centralize pricing expectations, inadvertently creating artificial price floors, promoting algorithmic collusion, or worsening housing affordability.
- **Speculative Misuse of Data:** Aggregating highly granular property profiles (combining precise location, floor level, room layout, and condition) could be exploited by institutional investors for aggressive, targeted real estate speculation.

# References

- **Scikit-learn Supervised Learning**
  - [https://scikit-learn.org/stable/supervised\\_learning.html](https://scikit-learn.org/stable/supervised_learning.html)
- **Stacking Regressor**
  - <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.StackingRegressor.html>
- **Kaggle dataset:**
  - <https://www.kaggle.com/datasets/luvathoms/portugal-real-estate-2024>